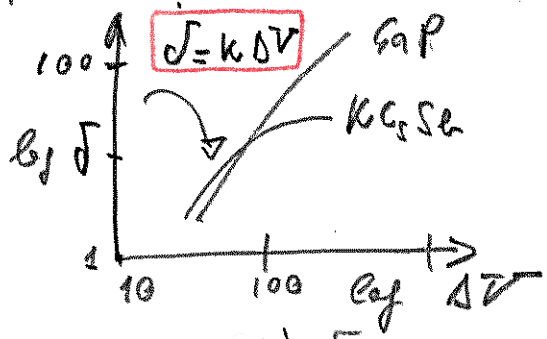
Photomultipliers

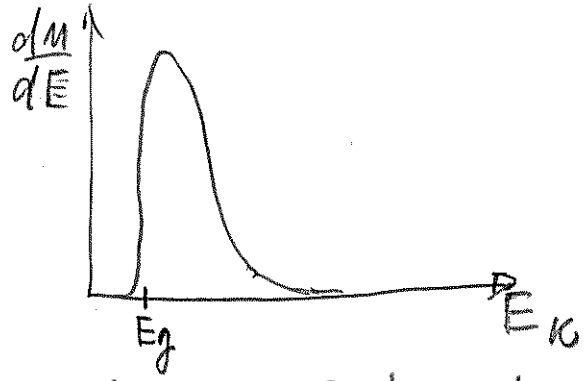
Different configurations for dynode arrangements

dynodes are made by antimony by alkaline metals, oxides (BeO, MgO) or semiconductors (GaP)

- δ secondary electrons are emitted on average per each incoming electron

δ depends on the accelerating potential





Gain for equal dynodes

$$G = \delta^N = K^N V_B^N = K V_B^N$$

Responsivity $[\Delta V \propto V_B]$

$$R_\lambda = G \eta \frac{e}{hc} \lambda$$

(N times larger than $\frac{\Delta V_B}{V_B}$) [ΔV = Voltage difference between dynodes]

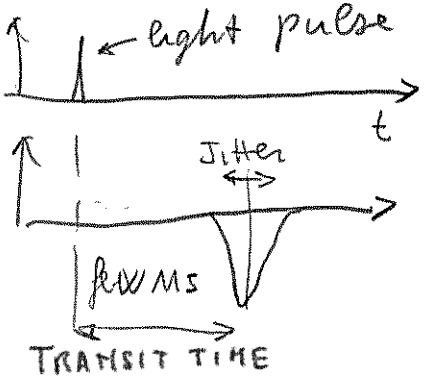
$$\frac{\Delta G}{G} = \frac{N K V_B^{N-1}}{K V_B^N} \Delta V_B = N \frac{\Delta V_B}{V_B}$$

← Sensitivity of the gain respect to supply voltage

To minimize gain variation the bias network has to be modified with zener diodes

This affects linearity

Time response



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Noise in PMTs

Dark current $\longrightarrow$ $J_T = \alpha T^2 \exp\left(-\frac{e\phi}{kT}\right)$ $\rightarrow$ Shot noise $i_S^{RMS} = \sqrt{\langle i_S^2 \rangle} = \sqrt{2eI_d\Delta f}$

$\uparrow$ cathode

Thermal noise $\longrightarrow$ $i_T^{RMS} = \sqrt{\langle i_T^2 \rangle} = \sqrt{4kT\Delta f / R_L}$

Variance of gain $\longrightarrow$ σ_G^2

$$RV_G = \frac{\sigma_G^2}{\langle G \rangle^2} = RV_1 + \frac{RV_2}{\delta_1} + \frac{RV_3}{\delta_1 \delta} + \dots + \frac{RV_n}{\delta_1 \delta^{n-2}}$$

[independent multipl. processes: $\delta_1, \delta \dots \delta$
δ = mult. factor of dynodes]

Assuming Poisson PDF for multipl. processes:

$$RV_G = \frac{1}{\delta_1} + \frac{1}{\delta_1 \delta} + \dots + \frac{1}{\delta_1 \delta^{n-1}}$$

$$RV_G = \frac{1}{\delta_1}\left(1 + \frac{1}{\delta} + \dots + \frac{1}{\delta^{n-1}}\right) = \frac{1}{\delta_1} \sum_{k=0}^{n-1} \frac{1}{\delta^k} = \frac{1}{\delta_1} \frac{\delta - \delta^{1-n}}{\delta - 1}$$

[Poisson PDF $\rightarrow \sigma^2 = \lambda = \delta$

$$RV_i = \frac{\sigma_\delta^2}{\langle \delta \rangle^2} = \frac{1}{\delta} \quad i = 1 \dots n$$

$$RV_G = \frac{1}{\delta_1} \frac{\delta - \cancel{\delta^{1-n}}}{\delta - 1} = \frac{1}{\delta_1} \frac{\delta}{\delta - 1} \quad \left[\text{if } \delta_1 = \delta \quad RV_G = \frac{1}{\delta - 1}\right]$$

$$\sum_{k=0}^{n} x^k = \frac{x^{n+1} - 1}{x - 1}$$

$$\sum_{k=0}^{n-1} x^k = \frac{x^{n+1} - 1}{x - 1} - x^n = \frac{x^n - 1}{x - 1}$$

Variance of electrons at cathode and Anode

$$RV_K = \frac{\sigma_{Ne}^2}{\langle Ne \rangle^2} = \frac{1}{\langle Ne \rangle} \quad \text{Poisson PDF}$$

$$RV_A = \frac{\sigma_A^2}{\langle N_A \rangle^2} = \frac{\langle Ne \rangle \langle G \rangle^2 + \sigma_G^2 \langle Ne \rangle}{\langle Ne \rangle^2 \langle G \rangle^2} = \frac{1}{\langle Ne \rangle} + \frac{1}{\langle Ne \rangle} \cdot RV_G$$

[σ_A^2 derives from two independent processes
1) STD of Ne $\times$ G
2) Variance of G $\times$ Ne

$$RV_A = RV_K \left(1 + RV_G\right)$$

$$NF = \frac{(S/N)^2_{IN}}{(S/N)^2_{OUT}} = \frac{RV_A}{RV_K} = 1 + RV_G = 1 + \frac{1}{\delta - 1} = \left[\frac{\delta}{\delta - 1}\right] \quad \delta \uparrow \; NF \downarrow$$

$$\left[NF = \frac{(S/N)^2_{IN}}{(S/N)^2_{OUT}} = \frac{\langle Ne \rangle^2 / \sigma_K^2}{\langle N_A \rangle^2 / \sigma_A^2} = \frac{1/RV_K}{1/RV_A}\right.$$

Noise in PMTs & NEP

For independent processes the variances must be summed

$$\sqrt{\langle i_s^2\rangle_{v.Ph.}} = \sqrt{2eI\Delta f} \xrightarrow{PMT} \sqrt{\langle i_s^2\rangle_A} = \sqrt{2(Ge)\cdot(GI)\cdot MF\cdot\Delta f} \qquad I = \cancel{I_d} + I_s$$

$$[S/N]_A = \frac{G I_s}{\sqrt{\langle i_s^2\rangle_A}} = \frac{G I_s}{G\sqrt{2eI_s MF\Delta f}} = \sqrt{\frac{I_s}{2e\,MF\,\Delta f}}$$ <u>WORSE</u> than at cathode

$$\sqrt{\langle i_T^2\rangle} = \sqrt{\frac{4KT\Delta f}{R_L}}$$

$$\sqrt{\langle i_{noise}^2\rangle} = \sqrt{\left(2e\cdot G^2\cdot I_s\cdot MF + \frac{4KT}{R_L}\right)\Delta f}$$

$$[S/N]_{output} = \sqrt{\frac{G^2 I_s^2}{\left(2eG^2 I_s MF + \frac{4KT}{R_L}\right)\Delta f}}$$

H. I_s:

$$\sqrt{\frac{G^2 I_s^2}{2eG^2 I_s MF\Delta f}} = \sqrt{\frac{I_s}{2e\,MF\,\Delta f}}$$ NOT CONVENIENT

L. I_s:

$$\sqrt{\frac{G^2 I_s^2}{\frac{4KT}{R_L}\Delta f}} = \frac{G I_s}{\sqrt{\frac{4KT}{R_L}\Delta f}} = G\cdot[S/N]_{v.Ph}$$ CONVENIENT

N.E.P.

$$NEP = \frac{\sqrt{\langle i_{noise}^2\rangle}}{R_\lambda} = \frac{1}{G}\frac{hc}{e\eta\lambda}\cdot\sqrt{\frac{4KT}{R_L}\Delta f} = \frac{[NEP]_{v.Ph.}}{G}$$ <u>Low I_{dark}</u>

$$NEP = \frac{1}{G}\frac{hc}{e\eta\lambda}\cdot\sqrt{\left(2e\,G^2 I_d\,MF + \frac{4KT}{R_L}\right)\Delta f} = \frac{hc}{e\eta\lambda}\cdot\sqrt{\left(2e\,I_d\,MF + \frac{4KT}{G^2 R_L}\right)\Delta f}$$

The contribution of Johnson noise is reduced by a factor G^2